

Name: KLOT Sopheak

ID:DUC2024-0042

Class: PG-A

Home Work

១. របៀប Download និង Install Docker

១.១ លើ Windows

- តម្រូវការប្រព័ន្ធ (Prerequisites): ត្រូវបើក WSL 2 (Windows Subsystem for Linux) និងមុខងារ Virtualization នៅក្នុង BIOS/Task Manager។
- ជំហានតម្លើង:
 1. ចូលទៅកាន់វេបសាយផ្លូវការ: docker.com/products/docker-desktop
 2. ចុចប៊ូតុង **Download for Windows** (ទទួលបានឯកសារ Docker Desktop Installer.exe)។
 3. បើកដំណើរការ installer ហើយជ្រើសយកជម្រើស **Use WSL 2 instead of Hyper-V** (recommended)។
 4. រង់ចាំការដំឡើងចប់ បន្ទាប់មកចុច **Close and restart** កុំភ្លេចបិទបើកកុំព្យូទ័រ។
 5. ពេលកុំព្យូទ័របើកមកវិញ សូមបើកកម្មវិធី **Docker Desktop**។
 6. បើក PowerShell ឬ Command Prompt (cmd) ហើយវាយពាក្យបញ្ជាដើម្បីពិនិត្យ៖
 - `docker --version`

```
PS C:\Users\MSI CYBORG> docker --version
Docker version 29.7.2, build a7dcaa6
PS C:\Users\MSI CYBORG> |
```

១.២ លើ Linux (Ubuntu/Debian) នៅលើ Linux យើងតម្លើងតាមរយៈ Command Line (Terminal) ដោយប្រើ APT repository៖

1. លុបកញ្ចប់ចាស់ៗចោល (ប្រសិនបើមាន):

- `sudo apt remove docker docker-engine docker.io containerd runc`

```
pheak@debian:~$ sudo apt remove docker docker-engine docker
io containerd runc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

2. អាប់ដេតប្រព័ន្ធ និងតម្លើង tools ចាំបាច់:

- `sudo apt update`
- `sudo apt install -y ca-certificates curl gnupg lsb-release`

```
pheak@debian:~$ sudo apt update
sudo apt install -y ca-certificates curl gnupg lsb-release
Hit:1 http://deb.debian.org/debian bookworm InRelease
Get:2 http://security.debian.org/debian-security bookworm-se
curity InRelease [34.8 kB]
Get:3 http://deb.debian.org/debian bookworm-updates InReleas
e [55.4 kB]
Hit:4 https://packages.sury.org/php bookworm InRelease

Get:5 http://security.debian.org/debian-security bookworm-se
curity/main Sources [250 kB]
Get:6 http://security.debian.org/debian-security bookworm-se
curity/main amd64 Packages [339 kB]
Get:7 http://security.debian.org/debian-security bookworm-se
curity/main Translation-en [212 kB]
Fetched 891 kB in 2s (399 kB/s)

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
266 packages can be upgraded. Run 'apt list --upgradable' to
see them.
Reading package lists... Done
```

3. បន្ថែម Docker Official GPG Key:

- `sudo install -m 0755 -d /etc/apt/keyrings`
- `curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg`
- `sudo chmod a+r /etc/apt/keyrings/docker.gpg`

```
pheak@debian:~$ sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo g
pg --dearmor -o /etc/apt/keyrings/docker.gpg
sudo chmod a+r /etc/apt/keyrings/docker.gpg
```

4. កំណត់ Repository:

- `echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg]
https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee
/etc/apt/sources.list.d/docker.list > /dev/null`

5. តម្លើង Docker Engine:

- `sudo apt update`
- `sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin`

6. អនុញ្ញាតឱ្យ user ធម្មតាដំណើរការ docker បាន (មិនបាច់ប្រើ sudo រាល់ដង):

- `sudo usermod -aG docker $USER`
 - `newgrp docker`
- ```
pheak@debian:~$ sudo usermod -aG docker $USER
newgrp docker
```

#### 7. តេស្តមើល version:

- `docker --version`
- ```
pheak@debian:~$ docker --version  
Docker version 20.10.24+dfsg1, build 297e128
```

២. បង្កើតគណនី Docker, ទាញយក (Pull) និង បង្ហោះ (Push) Image

ជំហាននេះអនុវត្តដូចគ្នាទាំងលើ Windows (PowerShell) និង Linux (Terminal)។

២.១ បង្កើតគណនី Docker Hub

1. ចូលទៅកាន់ hub.docker.com រួចចុច Sign Up។
2. បំពេញព័ត៌មាន (Docker ID/Username, Email, Password)។ ចំណាំទុក Username របស់អ្នក។
3. ចូលទៅ Email របស់អ្នកដើម្បីធ្វើការ Confirm/Verify គណនី។

២.២ Login ចូល Docker Hub លើ Command Line វាយពាក្យបញ្ជាខាងក្រោម រួចបញ្ចូល Username និង Password របស់អ្នក៖

➤ docker login

```
pheak@debian:~$ docker login
Login with your Docker ID to push and pull images from Docker Hub. If you don't have a Docker ID, head over to https://hub.docker.com to create one.
Username: sopheak12
Password:
WARNING! Your password will be stored unencrypted in /home/pheak/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
```

(ពេលជោគជ័យ វានឹងបង្ហាញសារ: Login Succeeded)

1.ទាញយក Image គំរូ (Pull Image): ជំហានតេស្តទី ១.

ទាញយក Image តូចមួយ (Alpine Linux) មកកាន់ម៉ាស៊ីន Debian របស់អ្នក៖

➤ docker pull alpine

```
pheak@debian:~$ docker pull alpine
Using default tag: latest
latest: Pulling from library/alpine
55afa1ecc21d: Pull complete

Digest: sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b
Status: Downloaded newer image for alpine:latest
docker.io/library/alpine:latest
```

- ការផ្ទៀងផ្ទាត់៖ វាយបញ្ជា docker images ដើម្បីពិនិត្យ។ អ្នកនឹងឃើញ alpine ស្ថិតនៅក្នុងបញ្ជី។

2.ដាក់ Tag ជាមួយឈ្មោះ Docker ID របស់អ្នក៖ ជំហានត្រៀម Push.

ដើម្បីបង្ហាត់ទៅកាន់ឈ្មោះរបស់អ្នកបាន Image នោះត្រូវតែមានឈ្មោះចាប់ផ្តើមដោយ sopheak12/៖

- `docker tag alpine sopheak12/my-first-image:v1`

```
|pheak@debian:~$ docker tag alpine sopheak12/my-first-image:v1
```

- **ការផ្ទៀងផ្ទាត់៖** វាយ `docker images` ម្តងទៀត អ្នកនឹងឃើញ Image ថ្មីឈ្មោះ `sopheak12/my-first-image` ជាមួយ Tag `v1`។

3.បង្ហាត់: Image ទៅកាន់ Docker Hub (Push Image): ជំហានបញ្ចប់.

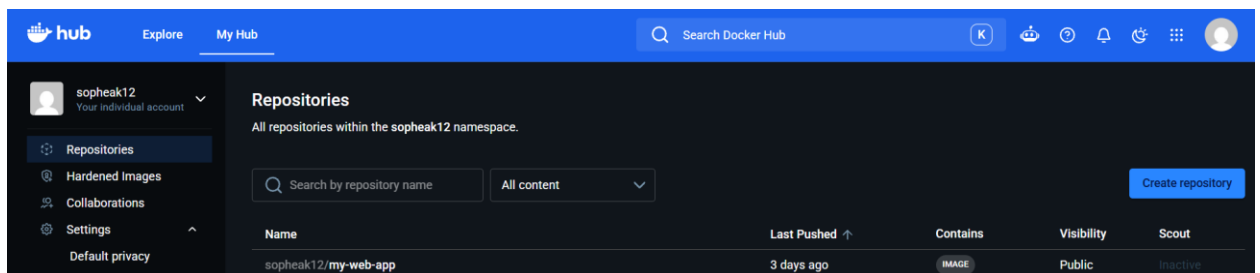
រុញ Image នោះឡើងទៅកាន់ Docker Hub Repository របស់អ្នក៖

- `docker push sopheak12/my-first-image:v1`

```
|pheak@debian:~$ docker push sopheak12/my-first-image:v1
The push refers to repository [docker.io/sopheak12/my-first-image]
34884abbe928: Mounted from library/alpine

v1: digest: sha256:96e9340afe8cd4ab5ee2989f5a544c3405ff2fc2450dd1a6f9eee2a63e6cda56 size: 527
```

- **ការផ្ទៀងផ្ទាត់៖** ចូលទៅកាន់ hub.docker.com ហើយពិនិត្យមើលក្នុងផ្ទាំង **Repositories** នោះអ្នកនឹងឃើញ `my-first-image` បង្ហាញនៅទីនោះ។



THANK YOU